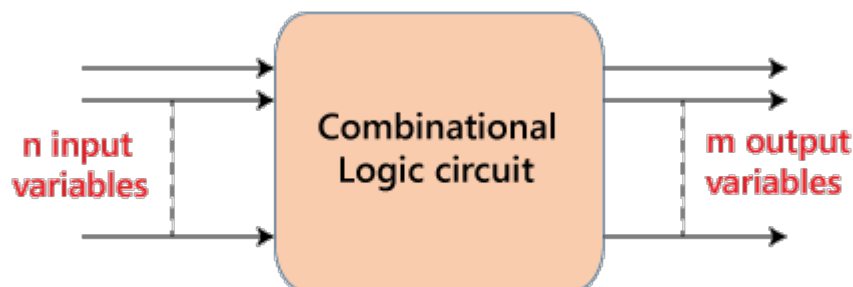


Combinational Logic circuits

The combinational logic circuits are the circuits that contain different types of logic gates. Simply, a circuit in which different types of logic gates are combined is known as a **combinational logic circuit**. The output of the combinational circuit is determined from the present combination of inputs, regardless of the previous input. The input variables, logic gates, and output variables are the basic components of the combinational logic circuit. There are different types of combinational logic circuits, such as Adder, Subtractor, Decoder, Encoder, Multiplexer, and De-multiplexer.

There are the following characteristics of the combinational logic circuit:

- o At any instant of time, the output of the combinational circuits depends only on the present input terminals.
- o The combinational circuit doesn't have any backup or previous memory. The present state of the circuit is not affected by the previous state of the input.
- o The n number of inputs and m number of outputs are possible in combinational logic circuits.



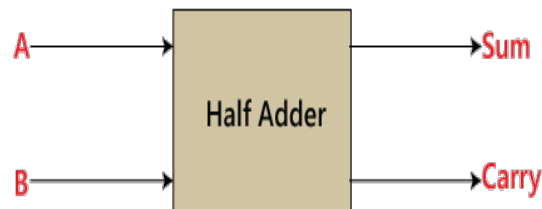
Block diagram of a combinational circuit

The ' n ' input variable comes from the external source while the ' m ' output variable goes to the external destination. In many applications, the source or destinations are storage registers.

Half Adder

The Half-Adder is a basic building block of adding two numbers as two inputs and produce out two outputs. The adder is used to perform OR operation of two single bit binary numbers. The **augend** and **addend** bits are two input states, and '**carry**' and '**sum**' are two output states of the half adder.

Block diagram



Half Adder Designing-

Half adder is designed in the following steps-

Step-01:

Identify the input and output variables-

- Input variables = A, B (either 0 or 1)
- Output variables = S, C where S = Sum and C = Carry

Step-02:

Truth Table

Inputs		Outputs	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

In the above table,

1. 'A' and 'B' are the input states, and 'sum' and 'carry' are the output states.
2. The carry output is 0 in case where both the inputs are not 1.

- The least significant bit of the sum is defined by the 'sum' bit.

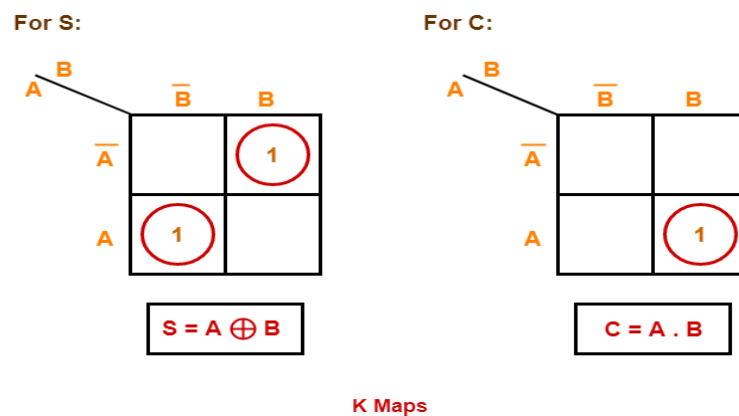
The SOP form of the sum and carry are as follows:

$$\text{Sum} = x'y + xy'$$

$$\text{Carry} = xy$$

Step-03:

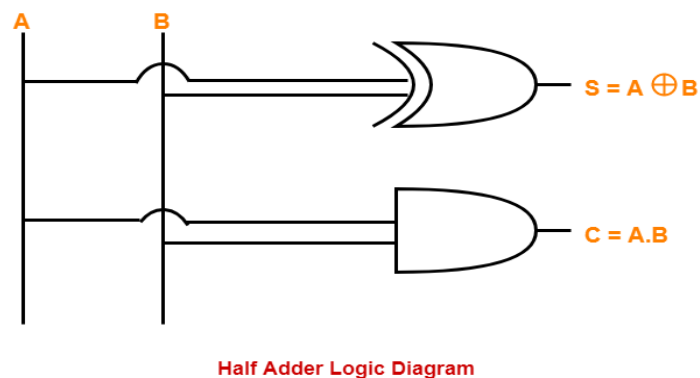
Draw K-maps using the above truth table and determine the simplified Boolean expressions-



Step-04:

Draw the logic diagram.

The implementation of half adder using 1 XOR gate and 1 AND gate is as shown below-



Limitation of Half Adder-

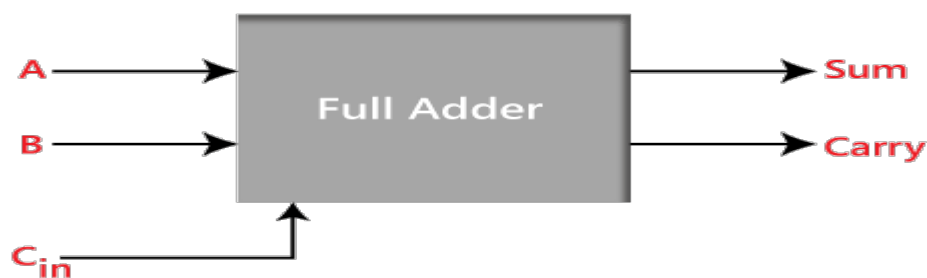
- Half adders have no scope of adding the carry bit resulting from the addition of previous bits.
- This is a major drawback of half adders.
- This is because real time scenarios involve adding the multiple number of bits which can not be accomplished using half adders.

To overcome this drawback, Full Adder comes into play.

Full Adder

The half adder is used to add only two numbers. To overcome this problem, the full adder was developed. The full adder is used to add three 1-bit binary numbers A, B, and carry C. The full adder has three input states and two output states i.e., sum and carry.

Block diagram



Truth Table

Inputs			Outputs	
A	B	C _{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

In the above table,

1. 'A' and 'B' are the input variables. These variables represent the two significant bits which are going to be added
2. 'C_{in}' is the third input which represents the carry. From the previous lower significant position, the carry bit is fetched.
3. The 'Sum' and 'Carry' are the output variables that define the output values.
4. The eight rows under the input variable designate all possible combinations of 0 and 1 that can occur in these variables.

Full Adder Designing-

Full adder is designed in the following steps-

Step-01:

Identify the input and output variables-

- Input variables = A, B, C_{in} (either 0 or 1)
- Output variables = S, C_{out} where S = Sum and C_{out} = Carry

Step-02:

Draw the truth table-

Inputs			Outputs	
A	B	C _{in}	S (Sum)	C _{out} (Carry)
0	0	0	0	0
0	0	1	1 _n	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1

1	1	1	1	1
---	---	---	---	---

Truth Table S=A

Step-03:

Draw K-maps using the above truth table and determine the simplified Boolean expressions-

For S:

	BC_{in}	$\overline{B}\overline{C}_{in}$	$\overline{B}C_{in}$	BC_{in}	$B\overline{C}_{in}$
$\overline{A}$		1			1
A	1			1	

$$S = A \oplus B \oplus C_{in}$$

For C_{out} :

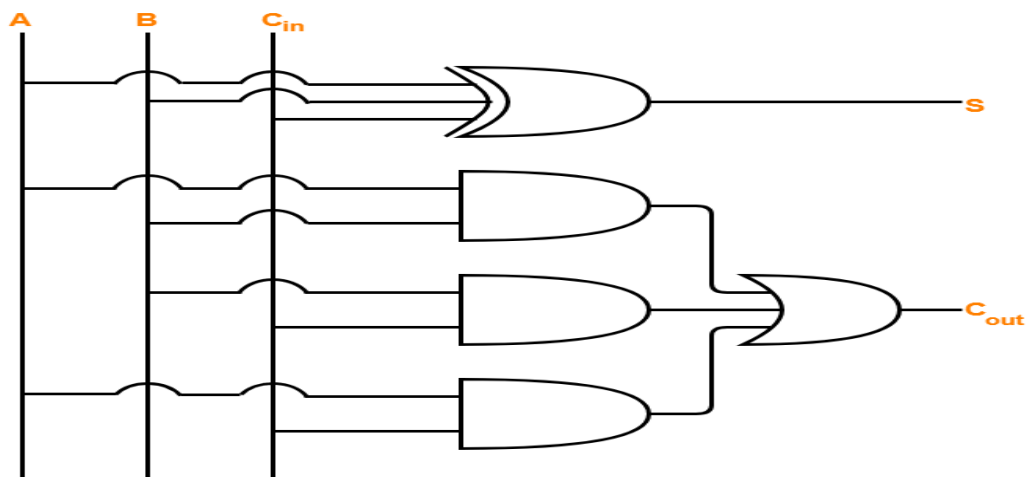
	BC_{in}	$\overline{B}\overline{C}_{in}$	$\overline{B}C_{in}$	BC_{in}	$B\overline{C}_{in}$
$\overline{A}$				1	
A		1		1	1

$$C_{out} = AB + BC_{in} + C_{in}A$$

Step-04:

Draw the logic diagram.

The implementation of full adder using 1 XOR gate, 3 AND gates and 1 OR gate is as shown below-

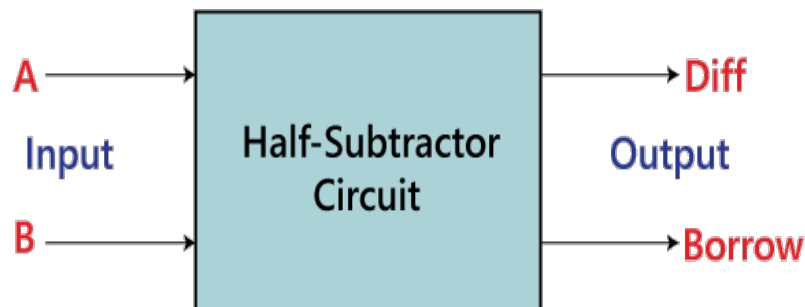


Full Adder Logic Diagram

Half Subtractor

The half subtractor is also a building block for subtracting two binary numbers. It has two inputs and two outputs. This circuit is used to subtract two single bit binary numbers A and B. The '**diff**' and '**borrow**' are two output states of the half subtractor.

Block diagram



Truth Table

Inputs		Outputs	
A	B	Diff	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

The SOP form of the **Diff** and **Borrow** is as follows:

$$\text{Diff} = A'B + AB'$$

$$\text{Borrow} = A'B$$

In the above table,

- o 'A' and 'B' are the input variables whose values are going to be subtracted.
- o The 'Diff' and 'Borrow' are the variables whose values define the subtraction result, i.e., difference and borrow.
- o The first two rows and the last row, the difference is 1, but the 'Borrow' variable is 0.
- o The third row is different from the remaining one. When we subtract the bit 1 from the bit 0, the borrow bit is produced.

Half Subtractor Designing-

Half subtractor is designed in the following steps-

Step-01:

Identify the input and output variables-

- Input variables = A, B (either 0 or 1)
- Output variables = D, b where D = Difference and b = borrow

Step-02:

Draw the truth table-

Inputs		Outputs	
A	B	D (Difference)	b (Borrow)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

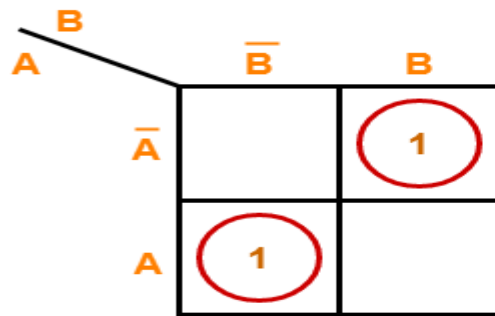
Truth Table

$$D=A'B+AB' \quad b=A'B$$

Step-03:

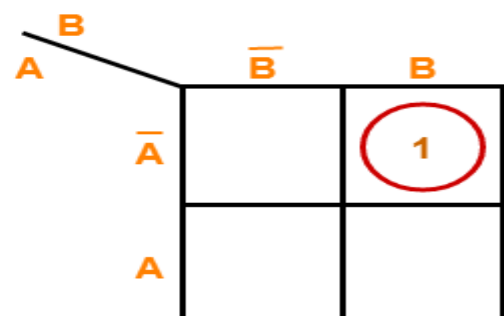
Draw K-maps using the above truth table and determine the simplified Boolean expressions-

For D:



$$D = A \oplus B$$

For b:



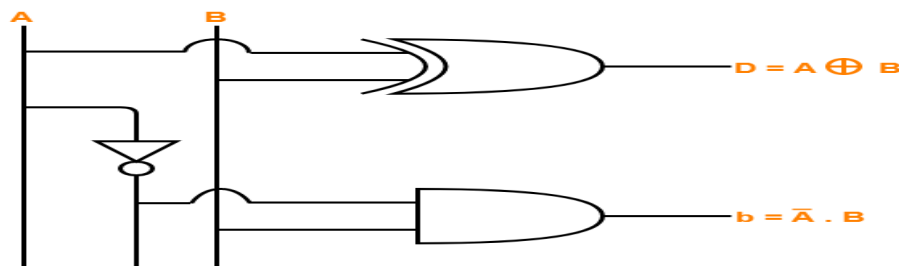
$$b = \bar{A}.B$$

K Maps

Step-04:

Draw the logic diagram.

The implementation of half subtractor using 1 XOR gate, 1 NOT gate and 1 AND gate is as shown below-



Half Subtractor Logic Diagram

Limitation of Half Subtractor-

- Half subtractors do not take into account "Borrow-in" from the previous circuit.
- This is a major drawback of half subtractors.
- This is because real time scenarios involve subtracting the multiple number of bits which can not be accomplished using half subtractors.